

SIERRA WIRELESS AIRPRIME EM7565 HARDWARE INTEGRATION MANUAL



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AirPrime EM7565

Hardware Integration Guide



SIERRA
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Proprietary and Confidential
Contents subject to change

Important Notice

Due to the nature of wireless communications, transmission and reception of data can never be guaranteed. Data may be delayed, corrupted (i.e., have errors) or be totally lost. Although significant delays or losses of data are rare when wireless devices such as the Sierra Wireless modem are used in a normal manner with a well-constructed network, the Sierra Wireless modem should not be used in situations where failure to transmit or receive data could result in damage of any kind to the user or any other party, including but not limited to personal injury, death, or loss of property. Sierra Wireless accepts no responsibility for damages of any kind resulting from delays or errors in data transmitted or received using the Sierra Wireless modem, or for failure of the Sierra Wireless modem to transmit or receive such data.

Safety and Hazards

Do not operate the Sierra Wireless modem in areas where blasting is in progress, where explosive atmospheres may be present, near medical equipment, near life support equipment, or any equipment which may be susceptible to any form of radio interference. In such areas, the Sierra Wireless modem **MUST BE POWERED OFF**. The Sierra Wireless modem can transmit signals that could interfere with this equipment.

Do not operate the Sierra Wireless modem in any aircraft, whether the aircraft is on the ground or in flight. In aircraft, the Sierra Wireless modem **MUST BE POWERED OFF**. When operating, the Sierra Wireless modem can transmit signals that could interfere with various onboard systems.

Note: Some airlines may permit the use of cellular phones while the aircraft is on the ground and the door is open. Sierra Wireless modems may be used at this time.

The driver or operator of any vehicle should not operate the Sierra Wireless modem while in control of a vehicle. Doing so will detract from the driver or operator's control and operation of that vehicle. In some states and provinces, operating such communications devices while in control of a vehicle is an offence.

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Sales information and technical support, including warranty and returns	Web: sierrawireless.com/company/contact-us/ Global toll-free number: 1-877-687-7795 6:00 am to 5:00 pm PST
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Revision History

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>> 1: Introduction

The Sierra Wireless EM7565 Embedded Module is an M.2 module that provides LTE, UMTS, and GNSS connectivity for notebook, ultrabook, tablet computers, and M2M applications over several radio frequency bands.

Accessories

A hardware development kit is available for AirPrime M.2 modules. The kit contains hardware components for evaluating and developing with the module, including:

- Development board
- Cables
- Antennas
- Other accessories

For over-the-air LTE testing, ensure that suitable antennas are used.

Required Connectors

[Table 1-1](#) describes the connectors used to integrate the EM7565 Embedded Module into your host device.

Table 1-1: Required Host–Module Connectors

Connector type	Description
RF cables	• Mate with M.2-spec connectors • Three connector jacks (I-PEX 20448-001R-081 or equivalent)
EDGE (67 pin)	• Slot B compatible—Per the M.2 standard (<i>[8] PCI Express NGFF (M.2) Electromechanical Specification Revision 1.0</i>), a generic 75 pin position EDGE connector on the motherboard uses a mechanical key to mate with the 67 pin notched module connector. • Manufacturers include LOTES (part #APCI0018-P001A01), Kyocera, JAE, Tyco, and Longwell.
SIM	• Industry-standard connector.

>> 2: Power

Power Supply

The host provides power to the EM7565 through multiple power and ground pins. The host must provide safe and continuous power (via battery or a regulated power supply) at all times; the module does not have an independent power supply, or protection circuits to guard against electrical issues.

For detailed pinout and voltage/current requirements of this module, see the *AirPrime EM7565 Product Technical Specification*.

Module Power States

The module has five power states, as described in [Table 2-1](#).

Table 2-1: Module Power States

State	Details	Host is powered	USB interface active	RF enabled
Normal (Default state)	- Module is active - Default state. Occurs when VCC is first applied, Full_Card_Power_Off# is deasserted (pulled high), and W_DISABLE# is deasserted - Module is capable of placing/receiving calls, or establishing data connections on the wireless network - Current consumption is affected by several factors, including: - Radio band being used - Transmit power - Receive gain settings - Data rate	✓	✓	✓
Low power ('Airplane mode')	- Module is active - Module enters this state: - Under host interface control: - Host issues AT+CFUN=0 (<i>AT Command Set for User Equipment (UE) (Release 6) (Doc# 3GPP TS 27.007)</i>), or - Host asserts W_DISABLE#, after AT!PCOFFEN=0 has been issued. - Automatically, when critical temperature or voltage trigger limits have been reached))	✓	✓	✗
Sleep	- Normal state of module between calls or data connections - Module cycles between wake (polling the network) and sleep, at network provider-determined interval.	✓	✗	✗

Table 2-1: Module Power States (Continued)

State	Details	Host is powered	USB interface active	RF enabled
Off	- Host keeps module powered off by asserting Full_Card_Power_Off# (signal pulled low or left floating) - Module draws minimal current	✓	✗	✗
Disconnected	Host power source is disconnected from the module and all voltages associated with the module are at 0 V.	✗	✗	✗

>> 3: RF Specifications

The EM7565 operates on the frequency bands listed below.

Table 3-1: LTE Frequency Band Support^a

Band	Frequency (Tx)	Frequency (Rx)
B1	1920–1980 MHz	2110–2170 MHz
B2	1850–1910 MHz	1930–1990 MHz
B3	1710–1785	1805–1880 MHz
B4	1710–1755	2110–2155 MHz
B5	824–849 MHz	869–894 MHz
B7	2500–2570 MHz	2620–2690 MHz
B8	880–915 MHz	925–960 MHz
B9	1749.9–1784.9 MHz	1844.9–1879.9 MHz
B12	699–716 MHz	729–746 MHz
B13	777–787 MHz	746–756 MHz
B18	815–830 MHz	860–875 MHz
B19	830–845 MHz	875–890 MHz
B20	832–862 MHz	791–821 MHz
B26	814–849 MHz	859–894 MHz
B28	703–748 MHz	758–803 MHz
B29	n/a	717–728 MHz
B30	2305–2315 MHz	2350–2360 MHz
B32	n/a	1452–1496 MHz
B41	2496–2690 MHz (TDD)	
B42	3400–3600 MHz (TDD)	
B43	3600–3800 MHz (TDD)	
B46	n/a	5150–5925 MHz (TDD)
B48	3550–3700 MHz (TDD)	
B66	1710–1780 MHz	2110–2200 MHz

a. For bandwidth support details, see *3GPP TS 36.521-1 v11.3.0, table 5.4.2.1-1*

Table 3-2: WCDMA Frequency Band Support

Band ^a	Frequency (Tx)	Frequency (Rx)
Band 1	1920–1980 MHz	2110–2170 MHz
Band 2	1850–1910 MHz	1930–1990 MHz
Band 3	1710–1785 MHz	1805–1880 MHz
Band 4	1710–1755 MHz	2110–2155 MHz
Band 5	824–849 MHz	869–894 MHz
Band 6	830–840 MHz	875–885 MHz
Band 8	880–915 MHz	925–960 MHz
Band 9	1749.9–1784.9 MHz	1844.9–1879.9 MHz
Band 19	830–845 MHz	875–890 MHz

a. WCDMA channel spacing is 5 MHz, but this can be adjusted to optimize performance in a particular deployment scenario.

Table 3-3: GNSS Frequency Band Support

Band	Frequencies
Narrow-band GPS, Galileo	Rx: 1575.42 MHz
Wide-band GPS + GLONASS	Rx: 1559–1606 MHz
Narrow-band BeiDou	Rx: 1561.098 MHz
Narrow-band GLONASS	Rx: 1601.72 MHz
Narrow-band QZSS	Rx: 1572.42 MHz

RF Connections

When attaching antennas to the module:

- Use RF plug connectors that are compatible with the following RF receptacle connectors: Foxconn (KK12011-02-7H), Longwell (911-002-0006R), Speedtech (C87P101-00001-H), Murata (MM4829-2702RA4 (HSC)), IPEX (20449-001E (MHF4)).
- Match coaxial connections between the module and the antenna to 50 Ω .
- Minimize RF cable losses to the antenna; the recommended maximum cable loss for antenna cabling is 0.5 dB.
- To ensure best thermal performance, use the mounting hole (if possible) to attach (ground) the device to a metal chassis.

Note: If the antenna connection is shorted or open, the modem will not sustain permanent damage.

Shielding

The module is fully shielded to protect against EMI and must not be removed.

Antenna and Cabling

When selecting the antenna and cable, it is critical to RF performance to match antenna gain and cable loss.

Choosing the Correct Antenna and Cabling

When matching antennas and cabling:

- The antenna (and associated circuitry) should have a nominal impedance of $50\ \Omega$ with a return loss of better than 10 dB across each frequency band of operation.
- The system gain value affects both radiated power *and* regulatory (FCC, IC, CE, etc.) test results.

Designing Custom Antennas

Consider the following points when designing custom antennas:

- A skilled RF engineer should do the development to ensure that the RF performance is maintained.
- If both UMTS and CDMA modules will be installed in the same platform, you may want to develop separate antennas for maximum performance.

Determining the Antenna's Location

When deciding where to put the antennas:

- Antenna location may affect RF performance. Although the module is shielded to prevent interference in most applications, the placement of the antenna is still very important—if the host device is insufficiently shielded, high levels of broadband or spurious noise can degrade the module's performance.
- Connecting cables between the module and the antenna must have $50\ \Omega$ impedance. If the impedance of the module is mismatched, RF performance is reduced significantly.
- Antenna cables should be routed, if possible, away from noise sources (switching power supplies, LCD assemblies, etc.). If the cables are near the noise sources, the noise may be coupled into the RF cable and into the antenna.

Disabling the Diversity Antenna

Use the AT command `!RXDEN=0` to disable receive diversity or `!RXDEN=1` to enable receive diversity.

Note: A diversity antenna is used to improve connection quality and reliability through redundancy. Because two antennas may experience difference interference effects (signal distortion, delay, etc.), when one antenna receives a degraded signal, the other may not be similarly affected.

Ground Connection

When connecting the module to system ground:

- Prevent noise leakage by establishing a very good ground connection to the module through the host connector.
- Connect to system ground using the module's mounting hole.
- Minimize ground noise leakage into the RF.

Depending on the host board design, noise could *potentially* be coupled to the module from the host board. This is mainly an issue for host designs that have signals traveling along the length of the module, or circuitry operating at both ends of the module interconnects.

Interference and Sensitivity

Several interference sources can affect the module's RF performance (RF desense). Common sources include power supply noise and device-generated RF.

RF desense can be addressed through a combination of mitigation techniques ([Methods to Mitigate Decreased Rx Performance on page 12](#)) and radiated sensitivity measurement ([Radiated Sensitivity Measurement on page 13](#)).

Note: The EM7565 is based on ZIF (Zero Intermediate Frequency) technologies. When performing EMC (Electromagnetic Compatibility) tests, there are no IF (Intermediate Frequency) components from the module to consider.

Interference From Other Wireless Devices

Wireless devices operating inside the host device can cause interference that affects the module.

To determine the most suitable locations for antennas on your host device, evaluate each wireless device's radio system, considering the following:

- Any harmonics, sub-harmonics, or cross-products of signals generated by wireless devices that fall in the module's Rx range may cause spurious response, resulting in decreased Rx performance.
- The Tx power and corresponding broadband noise of other wireless devices may overload or increase the noise floor of the module's receiver, resulting in Rx desense.

The severity of this interference depends on the closeness of the other antennas to the module's antenna. To determine suitable locations for each wireless device's antenna, thoroughly evaluate your host device's design.

Host-generated RF Interference

All electronic computing devices generate RF interference that can negatively affect the receive sensitivity of the module.

Proximity of host electronics to the antenna in wireless devices can contribute to decreased Rx performance. Components that are most likely to cause this include:

- Microprocessor and memory
- Display panel and display drivers
- Switching-mode power supplies

Device-generated RF Interference

The module can cause interference with other devices. Wireless devices such as AirPrime embedded modules transmit in bursts (pulse transients) for set durations (RF burst frequencies). Hearing aids and speakers convert these burst frequencies into audible frequencies, resulting in audible noise.

Methods to Mitigate Decreased Rx Performance

It is important to investigate sources of localized interference early in the design cycle. To reduce the effect of device-generated RF on Rx performance:

- Put the antenna as far as possible from sources of interference. The drawback is that the module may be less convenient to use.
- Shield the host device. The module itself is well shielded to avoid external interference. However, the antenna cannot be shielded for obvious reasons. In most instances, it is necessary to employ shielding on the components of the host device (such as the main processor and parallel bus) that have the highest RF emissions.
- Filter out unwanted high-order harmonic energy by using discrete filtering on low frequency lines.
- Form shielding layers around high-speed clock traces by using multi-layer PCBs.
- Route antenna cables away from noise sources.

Radiated Spurious Emissions (RSE)

When designing an antenna for use with AirPrime embedded modules, the host device with an AirPrime embedded module must satisfy any applicable standards/local regulatory bodies for radiated spurious emission (RSE) for receive-only mode and for transmit mode (transmitter is operating).

Note that antenna impedance affects radiated emissions, which must be compared against the conducted 50-ohm emissions baseline. (AirPrime embedded modules meet the 50-ohm conducted emissions requirement.)

Radiated Sensitivity Measurement

A wireless host device contains many noise sources that contribute to a reduction in Rx performance.

To determine the extent of any receiver performance desensitization due to self-generated noise in the host device, over-the-air (OTA) or radiated testing is required. This testing can be performed by Sierra Wireless or you can use your own OTA test chamber for in-house testing.

>> 4: Regulatory Compliance and Industry Certifications

This module is designed to meet, and upon commercial release, will meet the requirements of the following regulatory bodies and regulations, where applicable:

- Federal Communications Commission (FCC) of the United States
- The Certification and Engineering Bureau of Industry Canada (IC)
- The National Communications Commission (NCC) of Taiwan, Republic of China
- Ministry of Internal Affairs and Communications (MIC) of Japan
- Radio Equipment Directive (RED) of the European Union

Upon commercial release, the following industry certifications will have been obtained, where applicable:

- GCF
- PTCRB

Additional certifications and details on specific country approvals may be obtained upon customer request—contact your Sierra Wireless account representative for details.

Additional testing and certification may be required for the end product with an embedded EM7565 module and are the responsibility of the [OEM](#). Sierra Wireless offers professional services-based assistance to OEMs with the testing and certification process, if required.

Important Notice

Because of the nature of wireless communications, transmission and reception of data can never be guaranteed. Data may be delayed, corrupted (i.e., have errors) or be totally lost. Although significant delays or losses of data are rare when wireless devices such as the Sierra Wireless module are used in a normal manner with a well-constructed network, the Sierra Wireless module should not be used in situations where failure to transmit or receive data could result in damage of any kind to the user or any other party, including but not limited to personal injury, death, or loss of property. Sierra Wireless and its affiliates accept no responsibility for damages of any kind resulting from delays or errors in data transmitted or received using the Sierra Wireless module, or for failure of the Sierra Wireless module to transmit or receive such data.

Safety and Hazards

Do not operate your EM7565 module:

- In areas where blasting is in progress
- Where explosive atmospheres may be present including refuelling points, fuel depots, and chemical plants
- Near medical equipment, life support equipment, or any equipment which may be susceptible to any form of radio interference. In such areas, the EM7565 module **MUST BE POWERED OFF**. Otherwise, the EM7565 module can transmit signals that could interfere with this equipment.

In an aircraft, the EM7565 module **MUST BE POWERED OFF**. Otherwise, the EM7565 module can transmit signals that could interfere with various onboard systems and may be dangerous to the operation of the aircraft or disrupt the cellular network. Use of a cellular phone in an aircraft is illegal in some jurisdictions. Failure to observe this instruction may lead to suspension or denial of cellular telephone services to the offender, or legal action or both.

Some airlines may permit the use of cellular phones while the aircraft is on the ground and the door is open. The EM7565 module may be used normally at this time.

Important Compliance Information for United States and Canada

The EM7565 module, upon commercial release, will have been granted modular approval for mobile applications. Integrators may use the EM7565 module in their final products without additional FCC/IC (Industry Canada) certification if they meet the following conditions. Otherwise, additional FCC/IC approvals must be obtained.

1. At least 20 cm separation distance between the antenna and the user's body must be maintained at all times.
2. To comply with FCC/IC regulations limiting both maximum RF output power and human exposure to RF radiation, the maximum antenna gain including cable loss in a mobile-only exposure condition must not exceed the limits stipulated in [Table 4-1 on page 16](#).
3. The EM7565 module may transmit simultaneously with other collocated radio transmitters within a host device, provided the following conditions are met:
 - Each collocated radio transmitter has been certified by FCC/IC for mobile application.
 - At least 20 cm separation distance between the antennas of the collocated transmitters and the user's body must be maintained at all times.
 - The radiated power of a collocated transmitter must not exceed the EIRP limit stipulated in [Table 4-1 on page 16](#).

Table 4-1: Antenna Gain and Collocated Radio Transmitter Specifications

	Operating mode	Tx Freq Range (MHz)		Max Time-Avg Cond Power (dBm)	Antenna Gain Limit (dBi)		EIRP Limits (dBm)
					Standalone	Collocated	
EM7565 Embedded Module	WCDMA Band 2/ LTE B2	1850	1910	24	6	4	30
	WCDMA Band 4/ LTE B4	1710	1755	24	6	4	30
	WCDMA Band 5/ LTE B5	824	849	24	6	4	30
	LTE B7	2500	2570	23.8	9	4	32.8
	LTE B12	699	716	24	6	4	30
	LTE B13	777	787	24	6	4	30
	LTE B26	814	849	24	6	4	30
	LTE B30	2305	2315	23	1*	1*	32
	LTE B41	2496	2690	23.8	9	4	32.8
	LTE B48**	3550	3700	23	0	0	23
	LTE B66	1710	1780	24	6	4	30
Collocated transmitters	WLAN 2.4 GHz	2400	2500				30
	WLAN 5 GHz	5150	5850				30
	BT	2400	2500				16
	WiGig	58320	62640				25

***Important:** The FCC and IC have a strict EIRP limit in Band 30 for mobile and portable stations in order to protect adjacent satellite radio, aeronautical mobile telemetry, and deep space network operations. Mobile and portable stations must not have antenna gain exceeding 1 dBi in Band 30. Additionally, both the FCC and IC prohibit the use of external vehicle-mounted antennas for mobile and portable stations in this band.

Fixed stations may use antennas with higher gain in Band 30 due to relaxed EIRP limits. EM7565 modules used as fixed subscriber stations in Canada or fixed customer premises equipment (CPE) stations in the United States may have an antenna gain up to 10 dBi in Band 30, however, the use of outdoor antennas or outdoor station installations are prohibited except if professionally installed in locations that are at least 20 meters from roadways or in locations where it can be shown that the ground power level of -44 dBm per 5 MHz in the bands 2305–2315 MHz and 2350–2360 MHz or -55 dBm per 5 MHz in the bands 2315–2320 MHz and 2345–2350 MHz will not be exceeded at the nearest roadway. For the purposes of this notice, a roadway includes a highway, street, avenue, parkway, driveway, square, place, bridge, viaduct or trestle, any part of which is intended for use by the general public for the passage of vehicles.

Mobile carriers often have limits on total radiated power (TRP), which requires an efficient antenna. The end product with an embedded module must output sufficient power to meet the TRP requirement but not too much to exceed FCC/IC's EIRP limit. If you need assistance in meeting this requirement, please contact Sierra Wireless.

****Important:** Airborne operations in LTE Band 48 are prohibited.

4. A label must be affixed to the outside of the end product into which the EM7565 module is incorporated, with a statement similar to the following:
 - **This device contains FCC ID: N7NEM75**
Contains transmitter module IC: 2417C-EM75 where 2417C-EM75 is the module's certification number.
5. A user manual with the end product must clearly indicate the operating requirements and conditions that must be observed to ensure compliance with current FCC/IC RF exposure guidelines.

The end product with an embedded EM7565 module may also need to pass the FCC Part 15 unintentional emission testing requirements and be properly authorized per FCC Part 15.

Note: If this module is intended for use in a portable device, you are responsible for separate approval to satisfy the SAR requirements of FCC Part 2.1093 and IC RSS-102.

>> A: Acronyms

Table A-1: Acronyms and Definitions

Acronym or term	Definition
3GPP	3rd Generation Partnership Project
BeiDou	BeiDou Navigation Satellite System A Chinese system that uses a series of satellites in geostationary and middle earth orbits to provide navigational data.
BER	Bit Error Rate—A measure of receive sensitivity
BLER	Block Error Rate
dB	Decibel = $10 \times \log_{10} (P1/P2)$ <i>P1 is calculated power; P2 is reference power</i> Decibel = $20 \times \log_{10} (V1/V2)$ <i>V1 is calculated voltage, V2 is reference voltage</i>
dBm	A logarithmic (base 10) measure of relative power (dB for decibels); relative to milliwatts (m). A dBm value will be 30 units (1000 times) larger (less negative) than a dBW value, because of the difference in scale (milliwatts vs. watts).
DC-HSPA+	Dual Carrier HSPA+
EMC	Electromagnetic Compatibility
EMI	Electromagnetic Interference
FCC	Federal Communications Commission The U.S. federal agency that is responsible for interstate and foreign communications. The FCC regulates commercial and private radio spectrum management, sets rates for communications services, determines standards for equipment, and controls broadcast licensing. Consult www.fcc.gov .
Galileo	A European system that uses a series of satellites in middle earth orbit to provide navigational data.
GCF	Global Certification Forum
GLONASS	Global Navigation Satellite System—A Russian system that uses a series of 24 satellites in middle circular orbit to provide navigational data.
GNSS	Global Navigation Satellite Systems (GPS, GLONASS, BeiDou, and Galileo)
GPS	Global Positioning System An American system that uses a series of 24 satellites in middle circular orbit to provide navigational data.
Host	The device into which an embedded module is integrated
HSDPA	High Speed Downlink Packet Access
HSPA+	Enhanced HSPA, as defined in 3GPP Release 7 and beyond
HSUPA	High Speed Uplink Packet Access

Table A-1: Acronyms and Definitions (Continued)

Acronym or term	Definition
Hz	Hertz = 1 cycle/second
IC	Industry Canada
IF	Intermediate Frequency
IS	Interim Standard. After receiving industry consensus, the TIA forwards the standard to ANSI for approval.
LTE	Long Term Evolution—a high-performance air interface for cellular mobile communication systems.
MHz	Megahertz = 10e6 Hz
OEM	Original Equipment Manufacturer—a company that manufactures a product and sells it to a reseller.
OTA	'Over the air' (or radiated through the antenna)
PCB	Printed Circuit Board
PCS	Personal Communication System A cellular communication infrastructure that uses the 1.9 GHz radio spectrum.
PTCRB	PCS Type Certification Review Board
QZSS	Quasi-Zenith Satellite System—Japanese system for satellite-based augmentation of GPS.
RF	Radio Frequency
RSE	Radiated Spurious Emissions
Sensitivity (RF)	Measure of lowest power signal at the receiver input that can provide a prescribed BER/BLER/SNR value at the receiver output.
SNR	Signal-to-Noise Ratio
TIA/EIA	Telecommunications Industry Association / Electronics Industry Association. A standards setting trade organization, whose members provide communications and information technology products, systems, distribution services and professional services in the United States and around the world. Consult www.tiaonline.org .
UMTS	Universal Mobile Telecommunications System
USB	Universal Serial Bus
VCC	Supply voltage
WCDMA	Wideband Code Division Multiple Access (also referred to as UMTS)
WLAN	Wireless Local Area Network
ZIF	Zero Intermediate Frequency

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